Camtrap DP: Data standard for camera trap data

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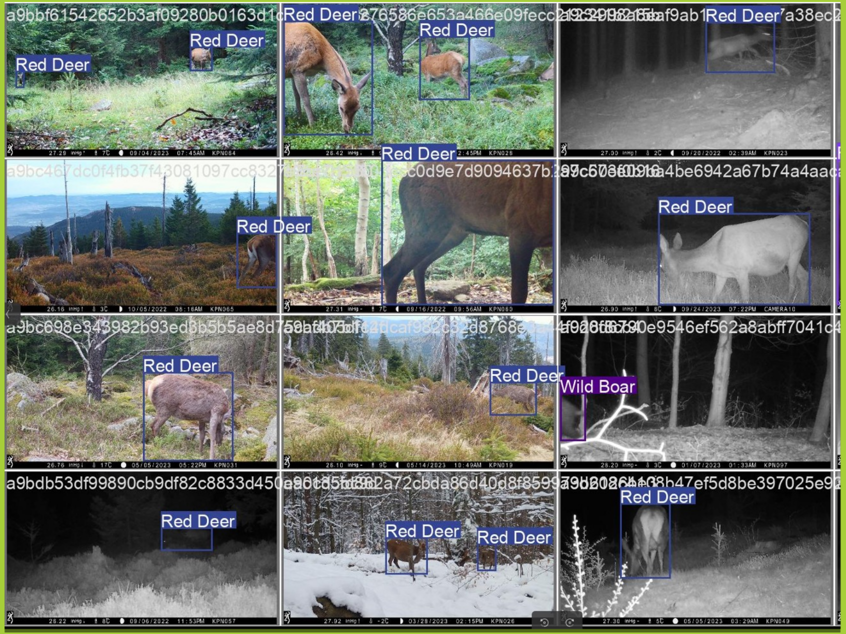
The role of Camtrap DP

- **Harmonizing** camera trapping data at a local-to-global scale
- Facilitating efficient and rapid **data exchange** and compilation
- Developing automated **AI-driven data processing pipelines**
- **Preserving** and **archiving** datasets with rich metadata
- <u>**Publishing to GBIF**</u>



Remote Sensing in Ecology and Conservation
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INTERDISCIPLINARY PERSPECTIVE

Camtrap DP: an open standard for the FAIR exchange and archiving of camera trap data

Jakub W. Bubnicki[1,2,*], Ben Norton[3], Steven J. Baskauf[4], Tom Bruce[5], Francesca Cagnacci[6,7], Jim Casaer[8], Marcin Churski[1,2], Joris P. G. M. Cromsigt[9], Simone Dal Farra[6], Christian Fiderer[10,11], Tavis D. Forrester[12], Heidi Hendry[13], Marco Heurich[10,11,14], Tim R. Hofmeester[9], Patrick A. Jansen[15], Roland Kays[3,16], Dries P. J. Kuijper[1], Yorick Liefting[15], John D. C. Linnell[17,18], Matthew S. Luskin[5], Christopher Mann[13], Tanja Milotic[8], Peggy Newman[19], Jürgen Niedballa[20], Damiano Oldoni[8], Federico Ossi[6], Tim Robertson[21], Francesco Rovero[22], Marcus Rowcliffe[23], Lorenzo Seidenari[24], Izabela Stachowicz[25,26], Dan Stowell[27,28], Mathias W. Tobler[29], John Wieczorek[30], Fridolin Zimmermann[31,32] & Peter Desmet[8,*]

TRAPPER

AGOUTI

CAMELOT

GBIF Global Biodiversity Information Facility

Atlas of Living Australia
ala.org.au
sharing biodiversity knowledge

Connection to other data standards

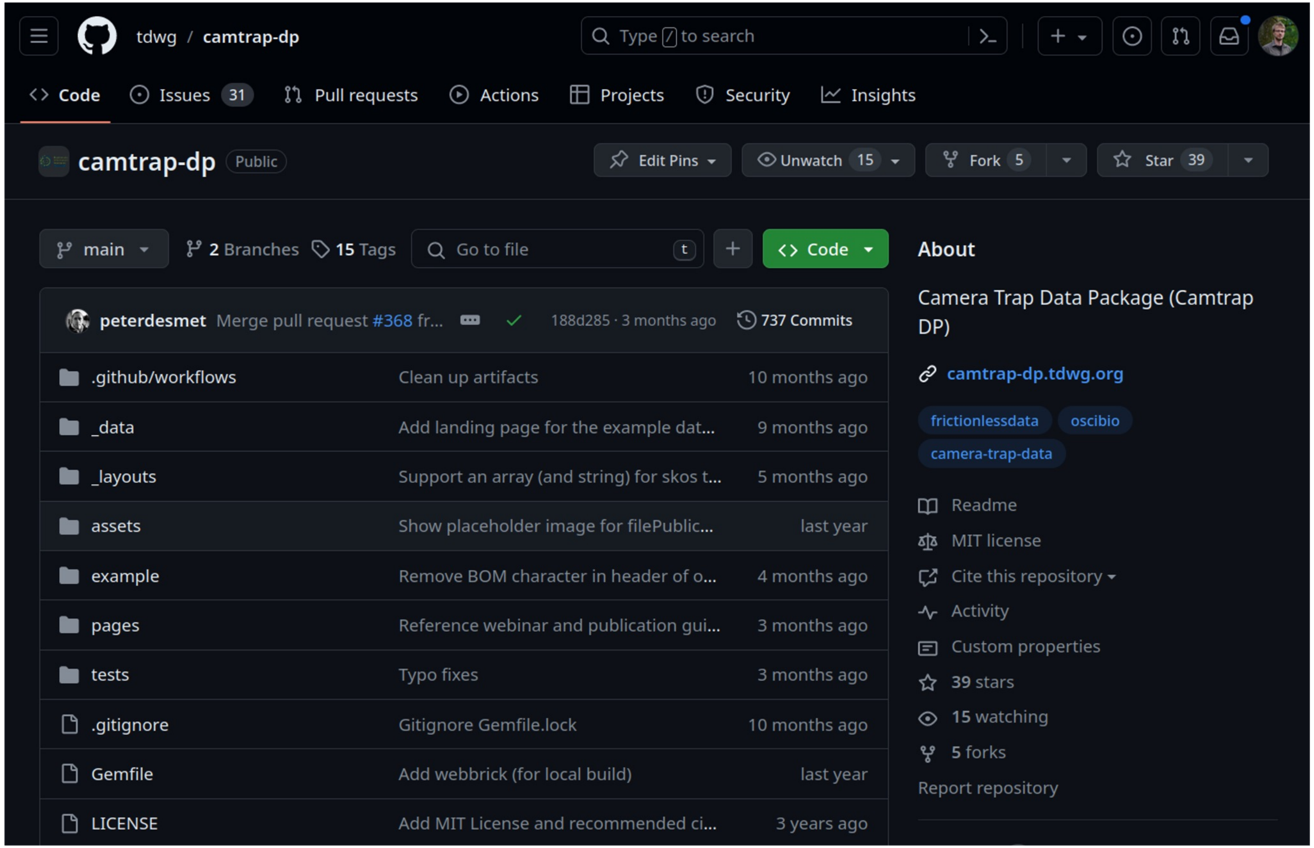
F indable
A ccessible
I nteroperable
R eusable

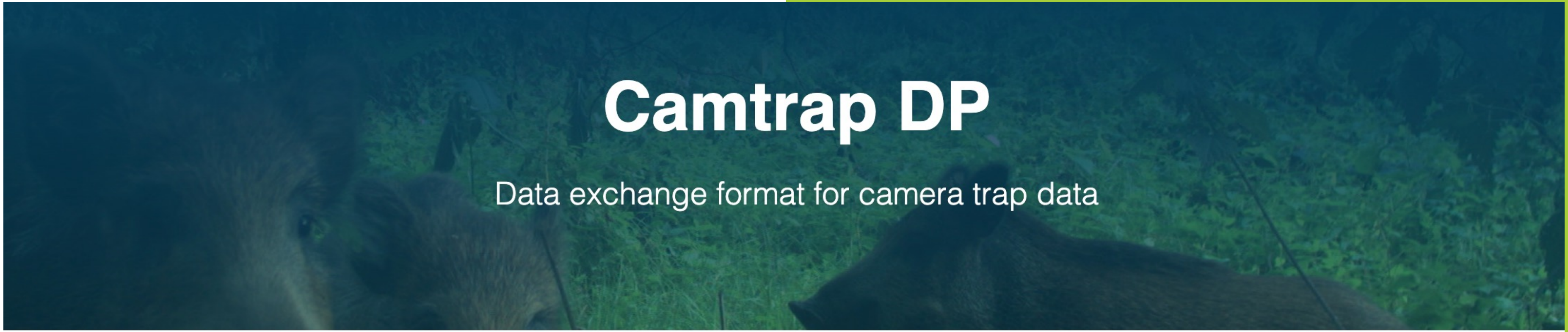
Biodiversity Information Standards
TDWG

Darwin Core

Frictionless Data

https://github.com/tdwg/camtrap-dp





Camera Trap Data Package (Camtrap DP) is a community-developed data exchange format for camera trap data. A Camtrap DP is a Frictionless Data Package that consists of:

File	Description
`datapackage.json`	Metadata about the data package and camera trap project.
`deployments.csv`	Table with camera trap placements (deployments).
`media.csv`	Table with media files recorded during deployments.
`observations.csv`	Table with observations derived from the media files.

https://camtrap-dp.tdwg.org/

datapackage .json

metadata

- general info about data package:
  name, description, version, source, contributiors, keywords
- licences for dataset and media
- list of all associated resources and used schemas
- information about study methodology:
  sampling design, capture method, whether individuals are recognized
- taxonomic, temporal and geographic scope

deployments .csv

metadata

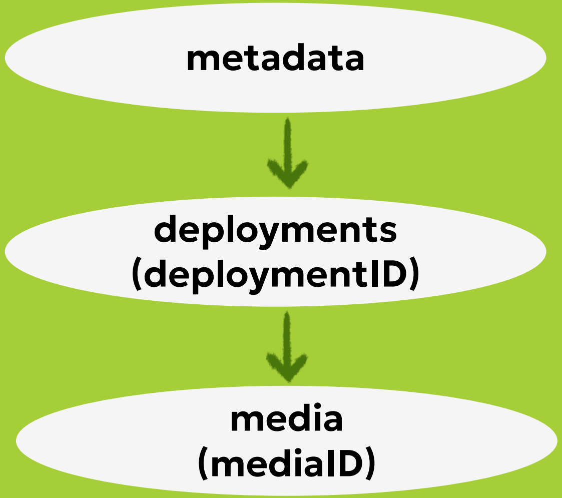
↓

deployments (deploymentID)

Deployment = placement of a camera trap at a given location, during some time period

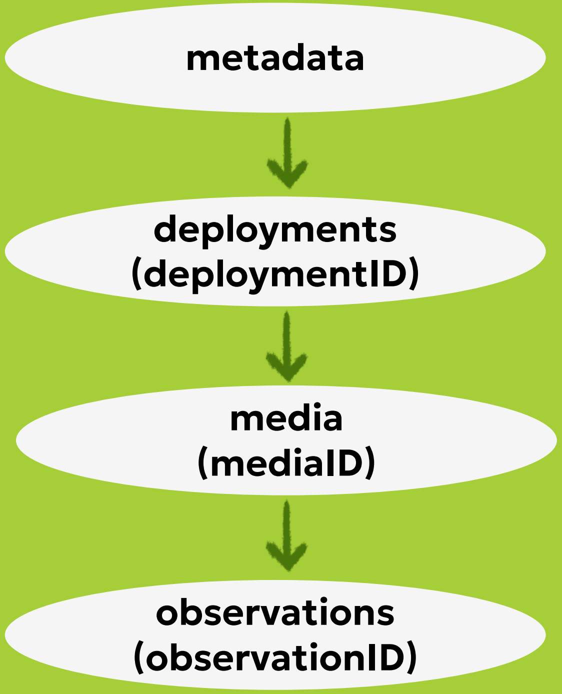
- location information
- start and end datetimes
- information about camera model and placement (height, depth, tilt, heading)
- extra info on deployment: bait use, habitat

media .csv



- connection to deployment
- filename
- how and where is the file accessible: public flag and file path
- media type (IANA)
- exif data (as json)

observations .csv



Consensus observations

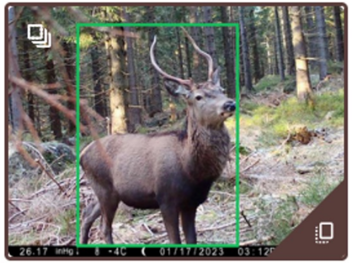
- connection to media
- eventID, start and end datetimes
- observation level (media or event)
- observation type (animal/human/vehicle/blank)
- info about the animal:

  taxon name, count, sex, behaviour, individual ID
- bbox coordinates
- classification info: who, how, when

Approved

IMG_0175

Approved



IMG_0176

Approved

IMG_0177

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IMG_0178

Approved

IMG_0179

media-based

- 5 observations with observationLevel=media
- don’t have to be mutually exclusive
- suitable for testing and training AI models

event-based

- 1 observation1 with observationLevel=event
- should be mutually exclusive
- suitable for ecological analyses

Helpful tools

camtrapdp

Camtrapdp is an R package to read and manipulate Camera Trap Data Packages (Camtrap DP). Camtrap DP is a data exchange format for camera trap data. With camtrapdp you can read, filter and transform data (including to Darwin Core) before further analysis in e.g. camtraptor or camtrapR.

https://inbo.github.io/camtrapdp/

frictionless-py

build passing | coverage 83% | pypi v5.19.0
DOI 10.5281/zenodo.15085933 | codebase github | support slack

Data management framework for Python that provides functionality to describe, extract, validate, and transform tabular data

https://framework.frictionlessdata.io/

Helpful tools

Biowatch

Biowatch for Windows | Biowatch for Mac | Biowatch for Linux

Analyze Camera Trap Data – Privately, On Your Machine

https://www.earthtoolsmaker.org/tools/biowatch/

Extensions of Camtrap DP

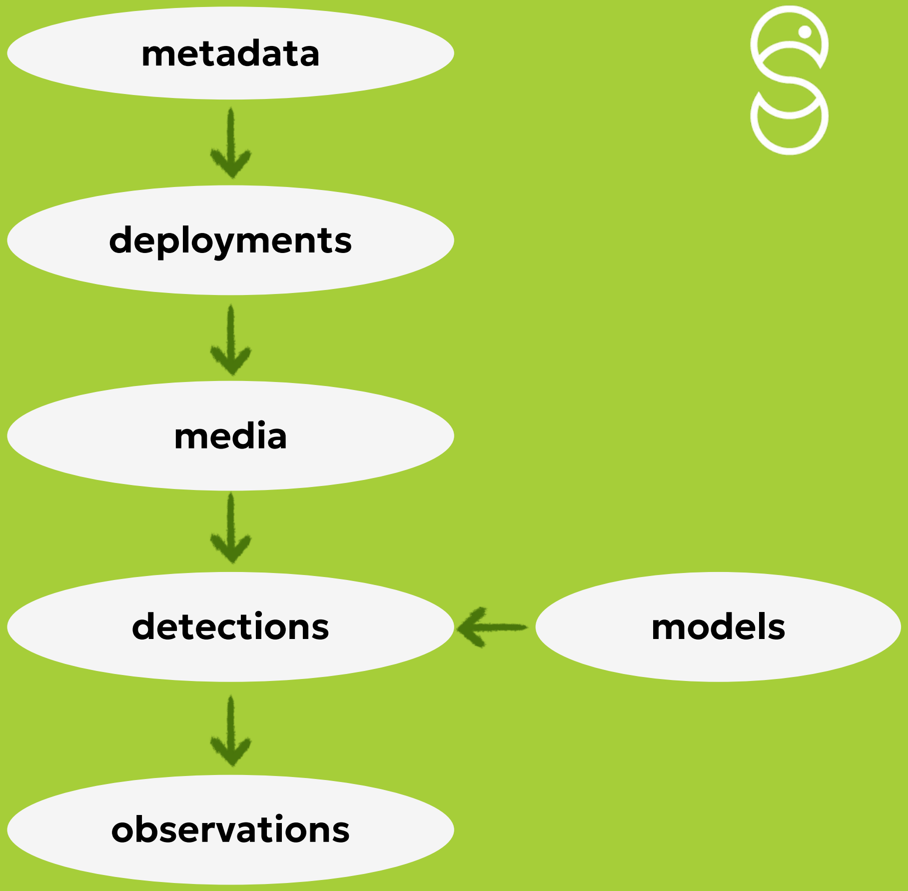
- Bioacoustics

Safe and Sound Project Report

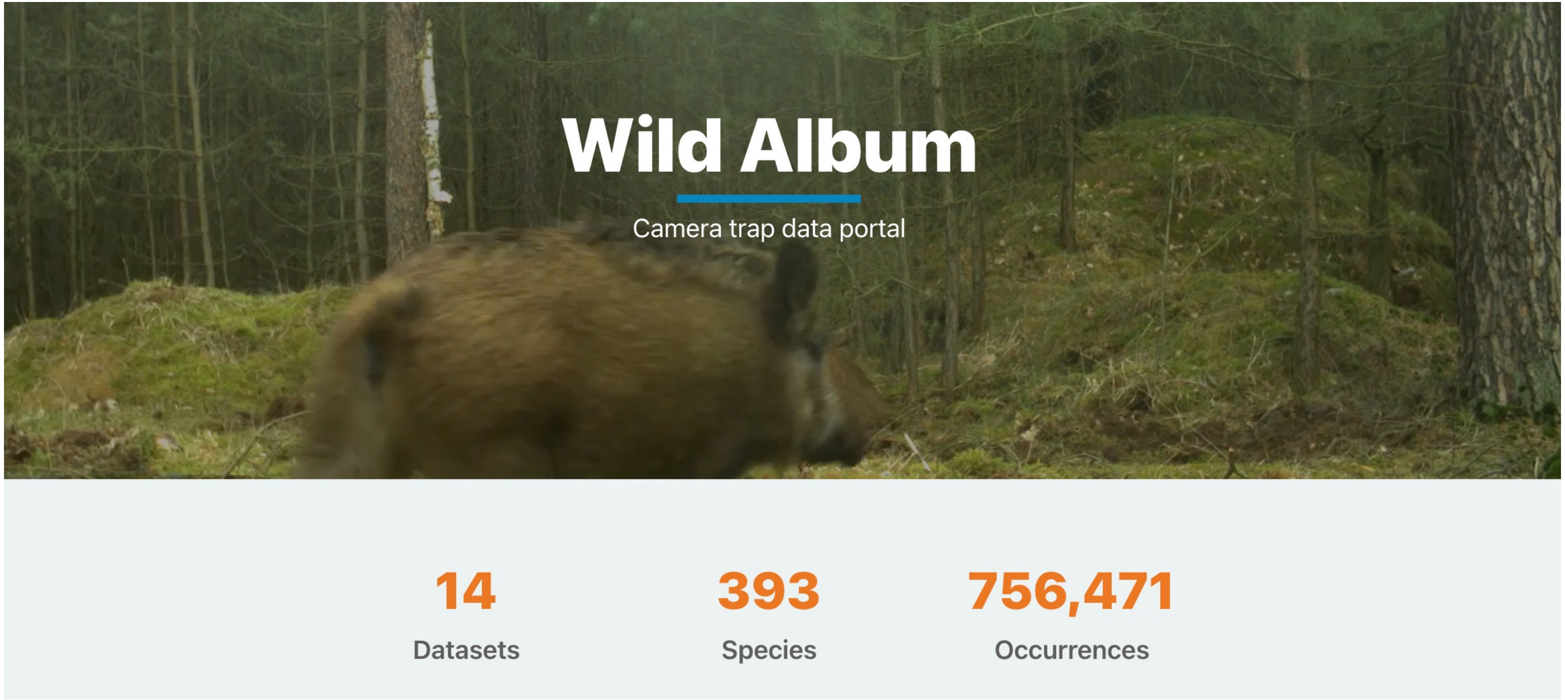
Is Camtrap DP a suitable standard for (bio)acoustic data?

Julia Wiel, Sanne Govaert, Benjamin Cretois, Peter Desmet

- Insect AI



Camtrap DP datasets published in GBIF



https://album.wildlabs.net/

OPEN SCIENCE CONSERVATION FUND

GBIF

InsectAI

Thank you!

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